

Charging the Corridors: A Distance-Weighted Difference-in-Differences Analysis of NEVI Interstate Fast Charging Deployment, Gasoline Consumption, and Air Quality across California Counties

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Introduction

- By 2023 California recorded ~1.26 million light-duty EV registrations and by late 2025 surpassed 200,000 public chargers, yet **gasoline use has not fallen proportionally** and transportation remains the state's largest source of greenhouse gas emissions.
- The 2022 **National Electric Vehicle Infrastructure (NEVI) Formula Program** directs federal funds to build DC fast-charging along designated Interstate Alternative-Fuel Corridors **before any other spending**.
- Because corridor placement is fixed by the federal interstate network and **state-administered rather than chosen by counties**, it offers a plausibly exogenous shock.
- Research question:** did the NEVI corridor build-out reduce per-capita gasoline consumption and improve local air quality in the counties it crosses, relative to comparable off-corridor counties?

Methodology

- Balanced California county-year panel:** 58 counties, 2010–2025 (928 county-years) from the CEC, U.S. DOE AFDC, EPA AQS, Census ACS, and the state NEVI corridor GIS layer.
- Treatment:** 23 corridor (treated) counties vs. 35 off-corridor (control), onset 2023; a county is treated if an interstate corridor crosses it ≥ 5 km.
- Two-way fixed-effects DiD** (county + year FE), controlling for log median household income and share driving alone; SEs clustered by county.
- Distance-weighted DiD** weights each county by centroid distance to the corridor — an exponential-decay kernel ($h = 50$ km) and inverse-distance ($1/d$) — concentrating identification near the threshold; event studies trace dynamics.
- Two outcomes:** log per-capita gasoline and median Air Quality Index (AQI):

$$Y_{ct} = \beta (\text{Treated}_c \times \text{Post}_t) + X_{ct}\gamma + \alpha_c + \delta_t + \varepsilon_{ct}$$

$$w_c = \exp(-\text{dist}_c/h) \quad \text{or} \quad w_c = 1/\text{dist}_c$$

Results

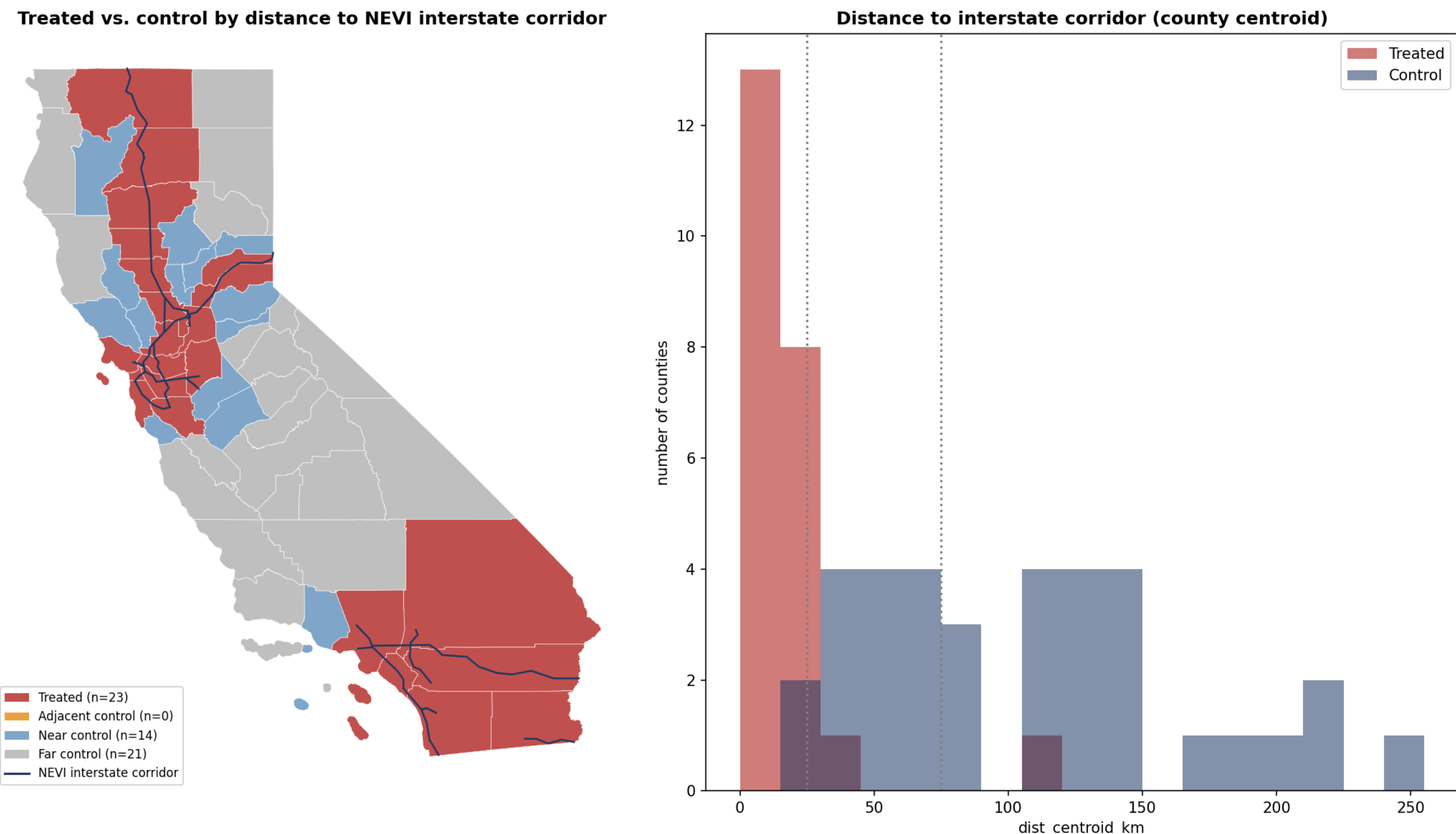


Figure 1: Treated versus control counties by distance to the NEVI interstate corridor (left) and the distribution of county-centroid distances (right).

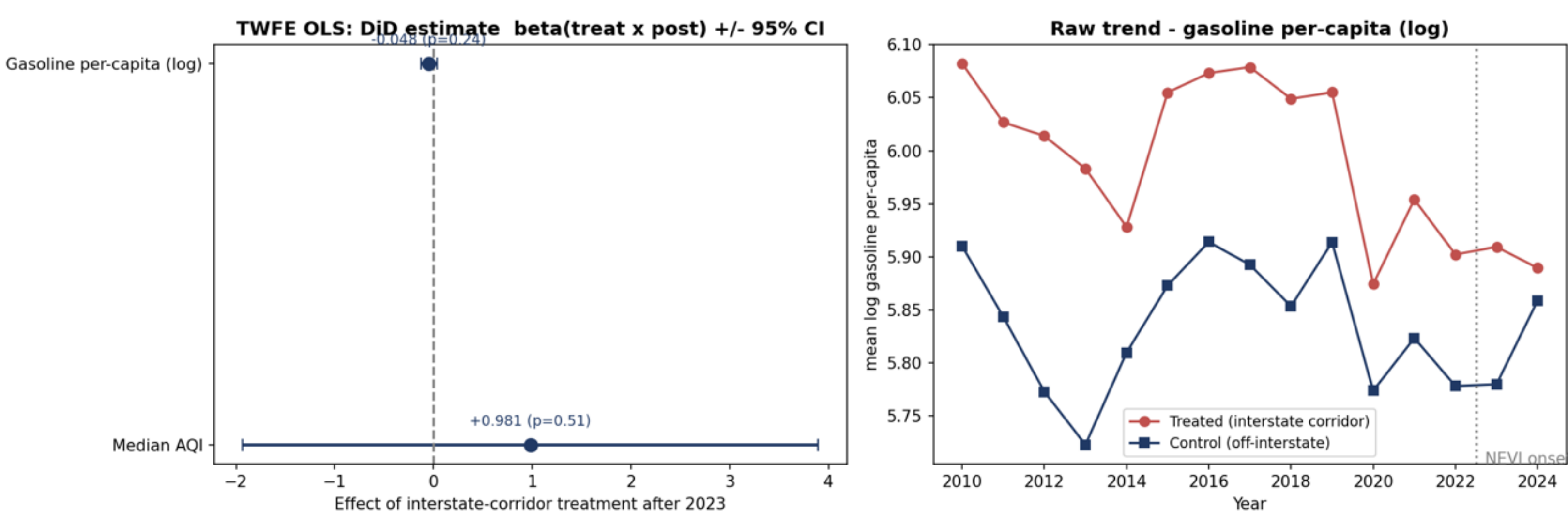


Figure 2: Baseline TWFE DiD estimate (left) and raw treated-vs-control gasoline trend (right).

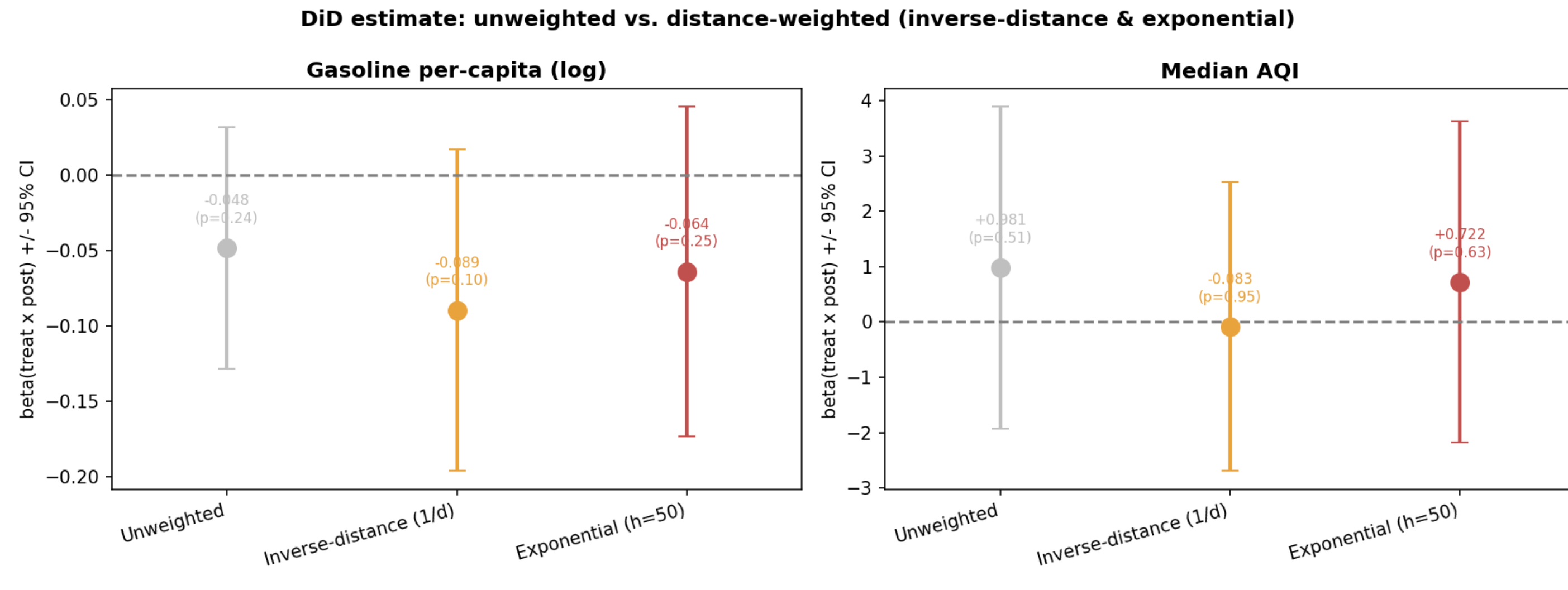


Figure 3: DiD under unweighted, inverse-distance, and exponential ($h = 50$ km) weighting.

Key Findings

- Baseline DiD on log gasoline is **-0.048** ($p = 0.24$); tilting weight toward corridor-adjacent counties makes it progressively more negative — **-0.064** (exponential, $h = 50$ km) and **-0.089** (inverse-distance, $p = 0.10$), roughly double the unweighted effect.
- Bandwidth surface:** the gasoline effect strengthens smoothly from -0.048 ($h = 250$ km) to **-0.130** ($h = 15$ km) — the more identification concentrates on corridor-adjacent counties, the larger the estimated fuel reduction.
- Event study:** flat, near-zero pre-trends, then the gasoline effect turns negative after the 2023 onset and reaches **-0.17 by 2024** ($p = 0.055$).
- No control county lies within 25 km** of the corridor; treated counties (mean ≈ 17 km) are separated from the nearest controls by a visible gap — exactly the spatial structure distance weighting exploits.
- Median AQI shows no detectable effect** under any weighting scheme over this short window.

Conclusion

- For gasoline the effect is **negative and grows stronger as weight concentrates on corridor-adjacent counties** ($-0.048 \rightarrow -0.089$), with an event-study estimate of ≈ -0.17 by 2024.
- This is **suggestive evidence that fuel displacement is concentrated near the state-built charging corridors** — where a proximity mechanism predicts — while stopping short of conventional significance.
- Median AQI shows no detectable corridor effect** over this short window.
- Main limitation: a **short post-treatment window** (gasoline observed only through 2024), so estimates are directional. Next steps: continuous-dose / staggered DiD and causal machine learning as more post-2023 data are collected.

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References

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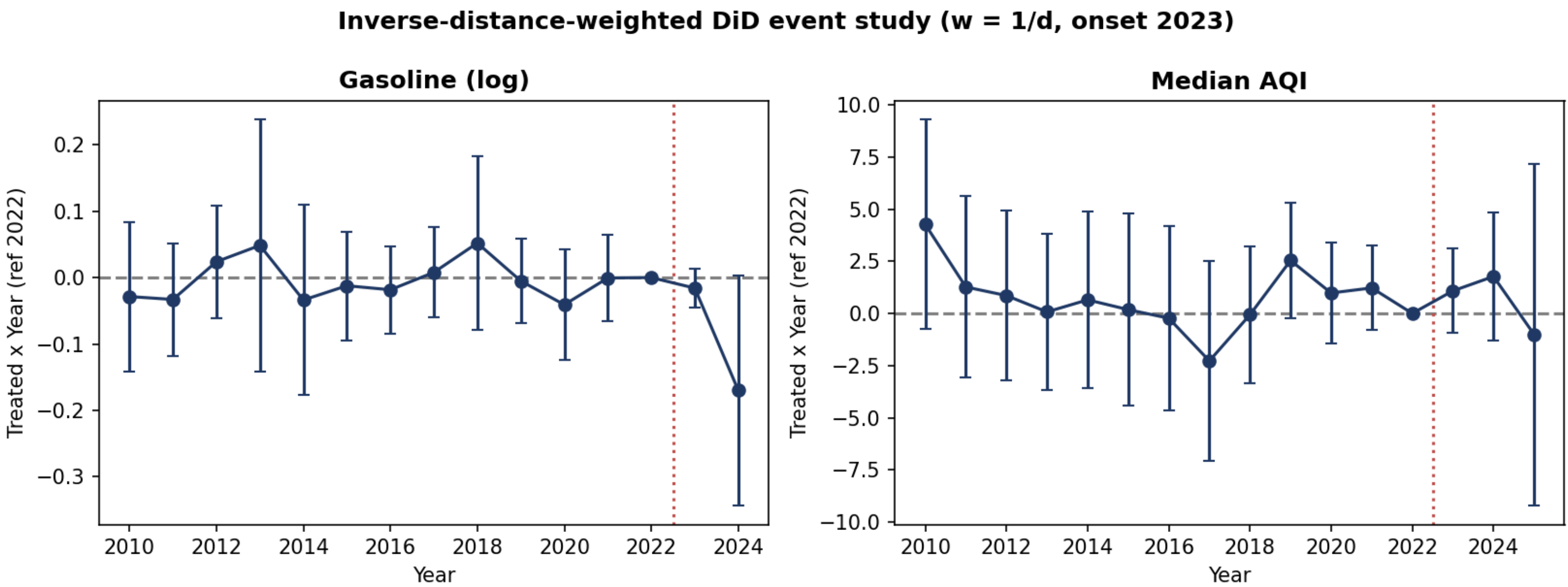


Figure 4: Inverse-distance-weighted DiD event study (reference 2022, onset 2023).